Trinominal nomenclature (*Tri* – three)

This naming system was proposed by Huxley and Stricklandt, Trinomen means, three names: generic name, species name and sub-species name. When members of any species which have large variations then trinomial system is used. On the basis of dissimilarities, this species gets classified into subspecies. It is the extension of binominal nomenclature system which has an addition of subspecies. All the three names are set in italics and only the first letter of generic name is capitalized, if handwritten then it should be underlined separately E.g.*Corvus splendens splendens* (Indian house crow)

Tautonymy: The practice of naming the animals in which the generic name and species name are the same, is called Tautonymy. e.g. *Naja naja* (The Indian Cobra).

What may be the reasons for the extinction of Dinosaurs? If you know the reasons for their extinction, why Sparrows are listed as endangered species?

Rules of Nomenclature

- The scientific name should be italicized in printed form and if handwritten, it should be underlined separately.
- The generic name's (*Genus*) first alphabet should be in uppercase.
- The specific name (*species*) should be in lowercase.
- The scientific names of any two organisms are not similar.
- The name or abbreviated name of the scientist who first publishes the scientific name may be written after the species name along with the year of publication. For example Lion-*Felis leo* Linn., 1758 or *Felis leo* L., 1758.
- If the species name is framed after any person's name the name of the species shall end with i, ii or ae.

For example, a new species of a ground-dwelling lizard (Cyrtodactylus) has been discovered and named after Scientist Varad Giri, *Cyrtodactylus varadgirii*.

1.7 Concept of species

Species is the basic unit of classification. The term species was coined by **John Ray**, and in his book ***"Historia Generalis Plantarum"*** (3 volumes) in 1693 described species as a group of morphologically similar organisms arising from a common ancestor. Carolus Linnaeus in his book ***"Systema naturae"*** considered species as the basic unit of classification. Species can be defined as a group of organisms that have similar morphology and physiology and can interbreed to produce fertile offsprings. In 1859 Charles Darwin in his book **Origin of species** explains the evolutionary connection of species by the process of natural selection.

1.8 Tools for study of taxonomy

Tools and taxonomical aids may be different for the study of plants and animals. Herbarium and Botanical garden

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